[INCH-POUND]

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COMMERCIAL ITEM DESCRIPTION

HAMMER, PILEDRIVER, SELF-POWERED, DIESEL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description covers self-powered, diesel piledriver hammers. The hammers are intended for use in driving wood, steel pipe, steel wide flange beam, concrete, and sheet piling.

2 CLASSIFICATION.

2.1 Classification. The hammers will be one of the following types and sizes, as specified (see 6.2):

Type I - For use with 20- to 22-inch leads.

Size A - 8,100 foot-pounds minimum rated striking energy
Size B - 16,000 foot-pounds minimum rated striking energy
Size C - 22,400 foot-pounds minimum rated striking energy
Size D - 8,000 foot-pounds minimum rated striking energy
Size E - 18,000 foot-pounds minimum rated striking energy

Type II - For use with 26-inch leads.

Size F - 30,000 foot-pounds minimum rated striking energy

Beneficial comments, recommendations, additions, deletions and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301.

AMSC N/A FSC 3895

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3. SALIENT CHARACTERISTICS.

3.1 Description. The hammer shall be self-contained, consisting essentially of a cylinder, free piston or ram, anvil, starting and stopping devices, fuel and lubricating oil systems, and throttle control. The self-powered hammer shall be air-cooled and shall operate on the two-stroke cycle compression-ignition principle. The hammer shall be furnished with lead guide, drive caps, leads, and bottom braces as specified (see 6.2).

3.2 Standard commercial product. The piledriver shall, as a minimum, be in accordance with the requirements of this commercial item description and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the piledriver being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market, through advertisements or manufacturer's catalogs or brochures, and represents the latest production model.

3.3 Safety. Reciprocating parts or components subjected to high temperatures, which are of such a nature or so located as to become a hazard to operating personnel, shall be enclosed, insulated, or guarded to the extent necessary to eliminate the hazard. This requirement does not apply to air-cooling fins. Protective devices shall not impair the operating functions of the hammer. A device shall be provided that shall prevent accidental engagement of the starting mechanism during hammer operations. The open-end cylinder-type hammer shall be equipped with a retainer to prevent the ram from jumping out of the cylinder on the upward stroke. A transport plug shall be furnished with each open-end cylinder-type hammer. Plug shall secure the ram within the cylinder.

3.4 Design. The hammer shall be designed to incorporate devices that shall protect moving parts from inclement weather and abrasive particles. A cushioning device shall be designed and furnished with the hammer that shall isolate excessive vibrations resulting from the ram striking the anvil during pile driving operations. Design of the hammer shall permit ready accessibility to all parts for maintenance and repair in the field by use of conventional general-purpose tools associated with hammers. The design of a single hammer may incorporate features that shall permit the use of more than one weight of ram in the same cylinder.

3.5 Operating weight. The operating weight shall include the complete hammer, lead guides, heaviest drive cap fillers, and adapters to be used. Each size of hammer shall conform to the requirements specified in table I.

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TABLE I. Operating weight, striking energy, and hp.

Type	Classi-fication size	Operating weight (lb) (max)	Ram weight (lb) (min)	Striking energy (ft-lb) (min)	Nominal Blow Rate full energy BPM (min)
I	A	5,500	1100	8,100	N/A
	B	7,500	2000	16,000	N/A
	C	10,500	2800	22,400	N/A
	D	6,000	1100	8,000	75
	E	13,000	2800	18,000	50
II	F	17,500	5000	30,000	N/A

3.6 Performance. The hammer shall operate in any ambient temperature from -25° Fahrenheit (F) through +125 °F. When operating with leads as specified herein, the hammer shall drive the piles specified in table II. 0

3.7 Lead guides. Unless otherwise specified (see 6.2), the hammer shall be furnished with one set of lead guides. The guides shall allow the hammer to operate with commercial standard box-type piledriver leads.

3.8 Drive caps. Each hammer shall be furnished with universal or separate drive caps as specified (see 6.2). The drive caps shall be suitable for use with commercial standard box leads and for driving the piles listed in table II. Overlap between each drive-cap guide and the lead shall be not less than 2 inches; the width between the inner faces of the leader and the width of the hammer jaws shall be ½" ± ¼ ". Hammers shall be furnished with a cushion block for inserting between the anvil and drive cap. The cushion block shall be either the disk type with laminated plastic and aluminum or the cast nylon type. The required type and quantity of cushion block to be furnished with each hammer shall be as specified (see 6.2).

TABLE II. Piles

Type of piling	Size of piling	Size of hammer					
		A	B	C	D	E	F
Timber	7 in. tip, 12 to 15 in. butt, 40 ft. long	X					
	Up to 15 in. butt, 9 in. tip and 10 x 10 in. square piling, 40 ft long				X		
	7 in. tip, 13 x 17 in. butt, 60 ft. long		X	X		X	

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TABLE II. Piles. (Continued)							
	Up to 18 in., 14 in. tip and square piling up to 14 x 14 in. size, 60 ft. long	A	B	C X	D	E X	F
	Up to 24 in., 18 in. tip and square piling up to 24 x 24 in. size, 60 ft. long						X
Stainless steel pipe	10 in. od, No. 7 gauge, 25 ft. long	X					
	Up to 10 in. od, No. 7 gauge, 25 ft. long 12 in. od, No. 7 gauge, 60 ft. long		X	X	X	X	
Concrete and steel-wide flange beam	8 in. and 10 in., 40 ft. long 10 in. and 12 in., 60 ft. long	X	X	X		X	
	12 in. and 14 in., 60 ft. long						X
Wide flange beam	8 in., 10 in., and 12 in., sizes, 40 ft. long			X		X	
	10 in., 12 in., sizes, 60 ft. long			X		X	
	12 in., 14 in., sizes, 60 ft. long						X
*Sheet steel pilling	3/8 in. web thickness, 40.5 lb. Lineal ft., 20 ft. long						
Steel pipe and round Concrete pile	Up to 14 in. dia., 60 ft. long			X X		X X	
	Up to 20 in. dial., 60 ft long						X

* Sheet steel piling conforming at ASTM A 328.

3.9 Fuel system The fuel system shall be complete with all operating accessories. The tank shall have sufficient capacity for not less than 8 hours of normal operation. The hammer shall operate on type and grade fuel as specified (see 6.2).

3.10 Lubricating system. The lubricating system shall be complete with an oil tank, filter, valves, pumps, hoses, and fittings. The oil tank shall have capacity to lubricate the hammer for not less than 8 hours of normal operation.

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3.11 Lifting device, tripping pendent line, and starting device. The hammer shall be equipped with tripping device. If the manufacturer's standard arrangement is designed so that the hammer can be lifted independent of the tripping pendent line when the tripping pendent line is connected to the starting device, the lifting lines or lifting chains conforming to OSHA 1910.184 and a tripping pendent line shall be furnished with the equipment. If the manufacturer's standard arrangement is designed so that the tripping device is used for lifting the hammer as well as for tripping the hammer with a single line, the tripping device design shall allow for easy removal and replacement of the crane's loadline. The starting device shall provide a means of lifting the ram to the top of the cylinder, and then releasing the ram automatically to begin its initial starting compression stroke. Engagement of the starting device shall be controlled manually. The hammer shall start in any ambient temperature from -25°F through +125°F. The use of priming agents as an extreme cold temperature initial starting aid is permissible. With pile set and ready for driving, the hammer shall start continuous firing, without the aid of ether or other priming methods, at normal ambient temperature above 32oF as follows:

3.12 Fuel control system. The hammer shall be furnished with a manual throttle control system which permits the energy output to be adjusted while the hammer is running. There shall also be a back-up fuel shut off method which is independent of the manual throttle. When specified (see 6.2), a remote fuel control system shall be furnished and powered hydraulically, pneumatically, or electrically. The remote control system shall permit the operator, when seated in the crane cab or standing on the ground, to stop the hammer or vary its performance from idle to full power. The remote system shall include 100 feet of control line in increments of 25 feet. When the remote control system is hydraulic- or pneumatic-powered, the control line shall be provided with self-sealing, quick-disconnect couplings. When electrically powered, the remote control system shall be completely waterproof. The control system shall allow for continuous operation of the hammer at any desired throttle setting. The hammer shall shut off automatically in case of failure of either control system.

3.13 Energy output gauge. The hammer shall be equipped with either a mechanical energy output gauge with visible indication of the energy output and energy output chart, or a mechanical stroke measurement gauge with visible indication of the stroke and a chart which correlates stroke to energy output. When specified (see 6.2), closed-end hammers shall be provided with a remote energy output gauge. The gauge shall provide a fast reading of the output energy during driving of the measured air pressure above the ram and in the compression chamber. A 50-foot hose shall be provided for connection of the gauge to the compression chamber with means of shutting off the hose at the gauge. A reference energy chart showing the equivalent weight-height (WH), stored, and potential energy in ft-lb, as obtained from the gauge dial reading, shall be furnished. The gauge and chart shall be enclosed in a metal box with overlapping cover not less than 16 gauge (0.0598 revised United States Standard gauge).

3.14 Leads Unless otherwise specified (see 6.2), one set of extended commercial standard box-type piledriver leads shall be furnished. The length of the leads shall require a 5-feet headroom clearance, plus normal operating length at maximum energy for the hammer and drive cap, plus the longest length of pile for the size hammer specified in table II. The leads shall be made in

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sections for bolting or pinning together. The leads shall be equipped with head blocks and movable boom point connectors with swivel and necessary sheaves and other accessories normally supplied to make a complete workable lead. The boom point connector shall be movable from not greater than 1/3 of the total lead length, or to a maximum of 25 feet. The boom point connector for type I hammer shall have a nominal 2.5-inch shaft, and for type II hammer a nominal 3-inch shaft. The lead sections shall be furnished with a ladder. The contractor shall show evidence that the leaders were designed using finite element analysis or other state of the art techniques used for the analysis of structures.

3.15 Bottom brace. A bottom brace shall be furnished with accessories for attachment to the crane and lead. The bottom brace shall be adjustable from a minimum length of not less than 15 feet to a maximum length of not less than 25 feet, with a minimum right and left side swing of 11 + 1° including angle from the centerline of the brace at minimum length. The bottom brace adjustment of length and swing shall be changed by mechanical means or hydraulic-driven means as specified (see 6.2). When the bottom brace is hydraulic-driven, means shall be provided to power the hydraulic system.

3.16 Instruction plates. The hammer shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.17 Identification marking. Identification shall be permanently and legibly marked directly on the hammer, or on a corrosion-resisting metal plate securely attached to the hammer at the source of manufacture. Identification shall include the manufacturer's model and serial number, name, energy output, ram weight, single or double acting, and trademark to be readily identifiable to the manufacturer. The identification plate shall be securely affixed to the hammer.

3.18 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

4. REGULATORY REQUIREMENTS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producers own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.3 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.4 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units. They must fall within the tolerances specified using conversion tables contained the latest revision of ASTM SI 10 (IEEE/ASTM SI 10). All other requirements of this Commercial Item Description including form, fit and function must also be met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in the contract or order.

5.1.1 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss and to accomplish reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.2 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate re-assembly. Parts and accessories removed and mating parts on the equipment shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The piledriver hammer with leads and bottom brace is intended to be used with a crane having a lattice-type boom for driving piles of the maximum length specified in table II for the size of hammer listed. Longer length piles may be driven by splicing a new section of pile onto the driven pile. However, the maximum driven pile length should not exceed a ram weight

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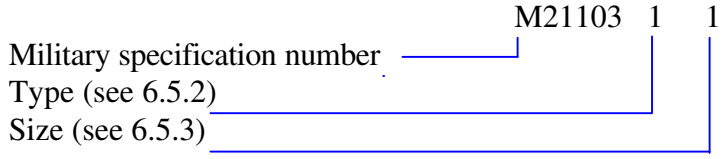
to pile weight ratio of 1 to 4. Fore and aft batter piles may be driven with a ratio of horizontal effect to vertical rise of 1:3; side by side batter piles may be driven with a ratio of 1:6..

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specifications.
b. Type, size of hammer required (see 1.2).
c. When lead guides are not required (see 3.7).
d. Type of drive caps required (see 3.8).
e. Type and quantity of cushion block required (see 3.8).
f. When a remote fuel control system is required (see 3.12).
g. When remote energy output gauge is required on hammer (see 3.13).
h. When leads are to be different (see 3.14).
i. When bottom brace is to be adjustable by mechanical means or hydraulic driven (see 3.15).
j. Type of fuel required for operation (see 3.9).

6.3 Part or identifying number (PIN). The PIN corresponds to the type and size data applicable to units covered by this specification and defines the requirements of the options presented under this specification. The military specification number and the type and size code numbers are combined to form the PIN.

6.3.1 Cataloging data. For cataloging purposes, PIN for the units are assigned as follows:



6.3.2 Type The type of the hammer (see 1.2) is identified by a single numerical character (see table III).

Table III. Code number to type

Type	Code
I	1
II	2

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6.3.3 Size. The size of the hammer (see 2.1) is identified by single alphabetic character (see table IV).

Table IV. Code letter to size.

Size	Code
A	1
B	2
C	3
D	4
E	5
F	6

6.4 ASTM Standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.5 IEEE Standards are available from the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017.

6.6 Subject term (key word) listing:
Construction equipment
Impactors

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD1

Review activity:
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FCAE

Preparing activity:
Navy YD1

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